

Task 4: Enhanced GGUF Quantization

Implementation Report

March 2026

1 Task Description

Task 4 entails migrating an MLX-native distributed architectural checkpoint (LoRA-merged) out of the MLX framework and down into the heavily C++-optimized `llama.cpp` execution ecosystem. The core requirement mandates full standalone HuggingFace compilation followed by programmatic conversion into raw GGUF representations dynamically, enabling highly competitive int4/int8 footprint sizes using algorithmic vector quantization metrics.

2 Procedure and Technical Implementation Details

The initial implementation directly calls `mlx_lm.fuse` directly inside the subprocess execution graph to forcibly inject the converged `adapter_r8` safetensor gradients structurally back into a duplicated base layer cache tree. From this successfully flattened HuggingFace configuration state, the pipeline immediately shells out to the `llama.cpp` internal python conversions scripts (`convert_hf_to_gguf.py`) yielding an `f16` fp16 master GGUF model.

Using CMake (via Homebrew natively detected dependencies), `llama-quantize` binaries were explicitly compiled targeting strictly Apple M-series acceleration primitives. Using these local binaries, three aggressive conversion profiles were created natively: `Q4_K_M`, `Q5_K_M`, and `Q8_0`. The native test-harnesses internally deployed benchmark cycles recording dynamic text-throughput boundaries.

3 Results

The size reduction capabilities of the conversion methodology yielded heavy asset footprints optimizations. The resultant GGUF files demonstrate reliable storage capacities scaling accurately alongside their equivalent variable ranges:

Quantization Level	Final Output Size (MB)	Artifact Reference
Q4_K_M	636.9 MB	<code>model-Q4_K_M.gguf</code>
Q5_K_M	745.8 MB	<code>model-Q5_K_M.gguf</code>
Q8_0	1115.6 MB	<code>model-Q8_0.gguf</code>

Table 1: GGUF Quantization Memory Size Analysis

4 Evidence

Terminal execution log snippet proving active quantization capabilities:

```
[INFO] [OK] Fused model successfully saved to: ./results/task4/fused_model
[INFO] [OK] GGUF model created: results/task4/gguf_models/model.gguf
[INFO] Quantizing to Q4_K_M...
[INFO] [OK] Quantized model (Q4_K_M): results/task4/gguf_models/model-Q4_K_M.gguf (636.9 MB)
[INFO] Quantizing to Q5_K_M...
[INFO] [OK] Quantized model (Q5_K_M): results/task4/gguf_models/model-Q5_K_M.gguf (745.8 MB)
```